

PYTHON · SQL SERVER · POWER BI · LOGISTICS ANALYTICS

Supply Chain Intelligence Analytics Platform

An end-to-end analytics project transforming 180,511 raw logistics transactions into actionable operational intelligence — from raw CSV through Python cleaning, SQL Server analysis, and Power BI executive dashboards.



Made with GAMMA

Dataset at a Glance

180,511

Clean Records

After removing 8 duplicate rows

35

Final Columns

After cleaning & feature engineering

\$36.78M

Total Revenue

Across all markets

\$3.97M

Total Profit

Net order-level profit

118

Unique Products

In the dataset

57.28%

Delayed Orders

Requiring operational attention

Structured tabular dataset covering Jan 2015 – Jan 2018, spanning 5 markets (Europe, LATAM, USCA, Africa, Pacific Asia) with numerical fields (sales, profit, shipping days) and categorical fields (market, region, department, shipping mode, product, segment, delivery status).

Business Problem

How can a supply chain business understand shipping delays, profitability patterns, product risk, and regional performance to improve operational efficiency?

Delivery Delay

Which shipping modes, products, and regions experience the highest delay rates?

Profitability

Which regions and departments generate revenue but struggle with margins?

Product Risk

Which products create high operational risk due to delayed deliveries?

Customer Segment

Which customer segments contribute the most revenue and profit?

Seasonal Trends

Which months generate peak sales and profit performance?

Executive Monitoring

How can management track KPIs through one Power BI dashboard?

Project Architecture & Analytics Flow



Raw data is transformed into cleaned features, SQL business intelligence outputs, and a Power BI dashboard — all documented in a final professional PDF report.

Python EDA & Cleaning Pipeline

01

Import & Inspect

Loaded supply_chain.csv; reviewed shape, dtypes, missing values, and duplicates.

02

Standardize & Clean

Converted headers to snake_case, removed redundant columns, stripped spaces, standardized categories.

03

Date-Time & Features

Converted order_date to datetime; extracted year, month, quarter, weekday.

04

Feature Engineering

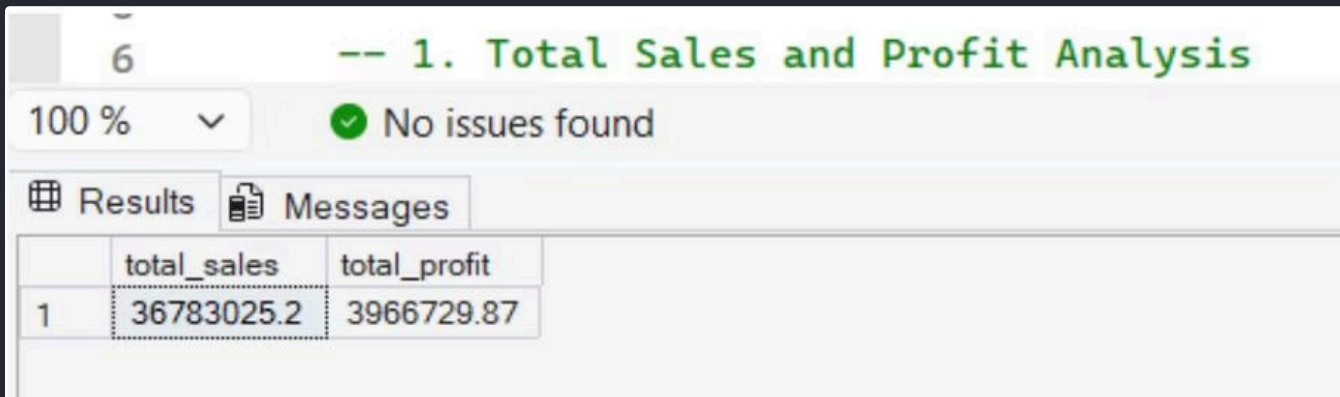
Created shipping_delay_days, delivery_performance, is_late_delivery, profit_margin_pct, revenue_per_unit, order_size, profit_type, sales_tier.

05

Validate & Export

Final shape: 180,511 rows × 35 columns, zero missing values, zero duplicates.

SQL Server Analysis – Key Findings



The screenshot shows a SQL Server Enterprise Manager interface. At the top, a query window displays the title "-- 1. Total Sales and Profit Analysis". Below the title, a status bar indicates "100 %" and "No issues found" with a green checkmark icon. The main area shows a table with two columns: "total_sales" and "total_profit". The first row of data shows a total sales of 36783025.2 and a total profit of 3966729.87.

	total_sales	total_profit
1	36783025.2	3966729.87

Revenue & Profit

Total sales: **\$36.78M** · Total profit: **\$3.97M**

Shipping Risk

Standard Class produced the largest late-delivery volume; +1 day was the most common delay pattern.

Top Categories

Fishing, Cleats, Camping & Hiking led revenue. Fan Shop dominated department-level sales (\$17.1M).

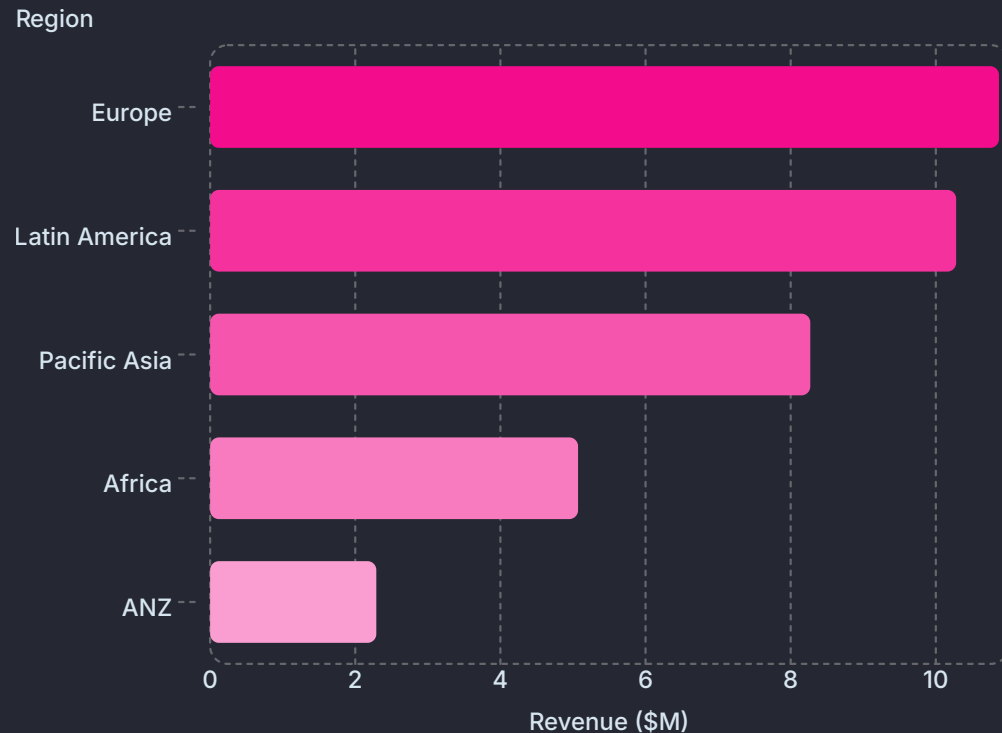
Customer Segment

Consumer segment generated highest revenue (\$19.09M) and order count (93,498).

High-Risk Products

Ogio Race Golf Shoes (68.85%) and Team Golf Putter Grip (65.18%) showed highest delay risk.

Regional & Market Performance



Key Regional Insights

- **Europe & LATAM** dominated total revenue contribution
- **Africa** achieved the highest profit margin at **11.23%** with consistent ~3-day delivery
- All markets averaged **3-day delivery**; Africa led operational efficiency
- Western Europe led regional sales at \$5.89M with 10.61% margin

Power BI Dashboard & KPI Reporting



Revenue & Profit Trend

Monthly line chart tracking revenue and profit from Jan 2015 to Mar 2018.

Revenue by Category

Donut chart across Electronics, Apparel, Home & Garden, and Beauty.

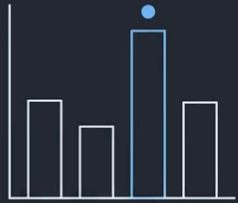
High-Risk Products

Top 10 products ranked by delivery delay risk — Ogio Race Golf leads at 76.44%.

Order Size Performance

Large orders: highest avg sales (\$349.58) and profit (\$27.06).

Key Insights



**FAN SHOP IS
TOP REVENUE &
PROFIT DEPT**
(\$17.1M SALES,
\$18.5M PROFIT)



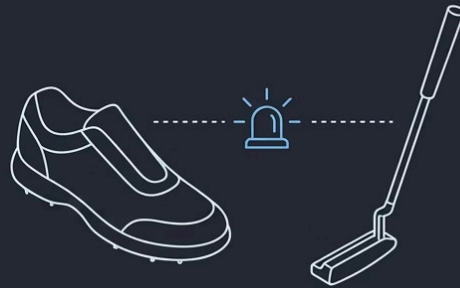
**AFRICA HAS
HIGHEST
PROFIT MARGIN**
(11.23%) WITH
~3-DAY DELIVERY



**DELAYED
ORDERS AT 57.28%
REQUIRE
OPERATIONAL
ATTENTION**



**LARGE ORDERS
GENERATE HIGHEST
AVG SALES (\$349.58)
AND PROFIT (\$27.06)**



**OGIO RACE GOLF SHOES &
TEAM GOLF PUTTER GRIP**
SHOW DELAY RISK ABOVE 65%

Operational Priorities

- Delayed deliveries carry **lower average profit** (\$21.59) vs on-time (\$22.53) and early (\$22.47)
- SOLE E35 Elliptical and SOLE E25 Elliptical generate **losses despite sales volume**
- Consumer segment drives **52% of total revenue** — highest-value customer group
- Peak months: **September 2017** led sales at \$1.14M

Conclusion & Future Scope

The project successfully transforms raw logistics data into operational intelligence using Python, SQL Server, and Power BI — highlighting delivery delay risks, profitable departments, regional performance, and product-level bottlenecks.

→ Live Shipment Tracking

Integrate real-time logistics monitoring for proactive delay management.

→ ML Delay Prediction

Predict late deliveries before dispatch using machine learning models.

→ Supplier Risk Scoring

Identify recurring fulfillment issues at the supplier level.

→ Power BI Service Deployment

Deploy dashboard with scheduled refresh and AI natural-language Q&A.

→ Automated Alerts

Trigger alerts for high delay percentages, negative profit, and rising shipping times.